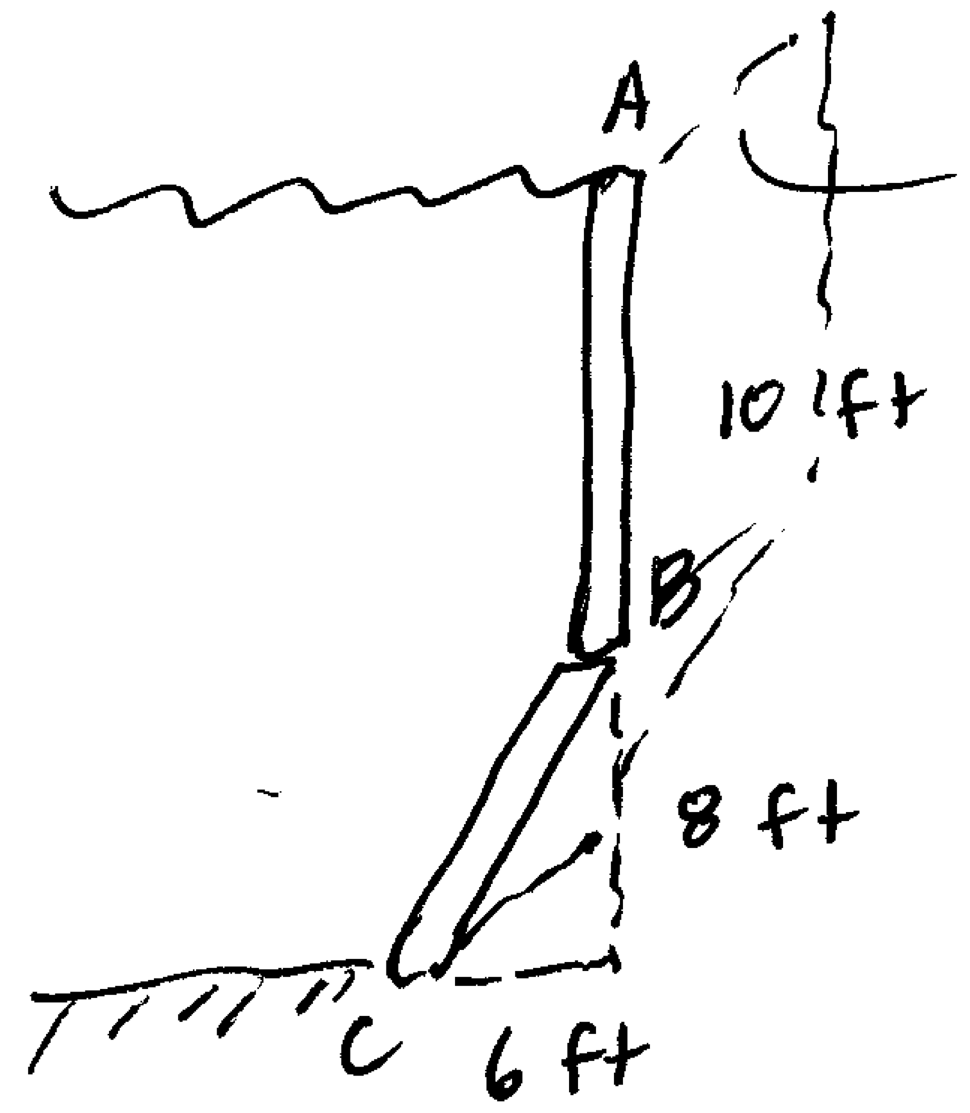
A third Fluid Pressure Problem worked three ways



$\gamma = 65 \ lb/ft^3$

(slightly heavier than fresh water

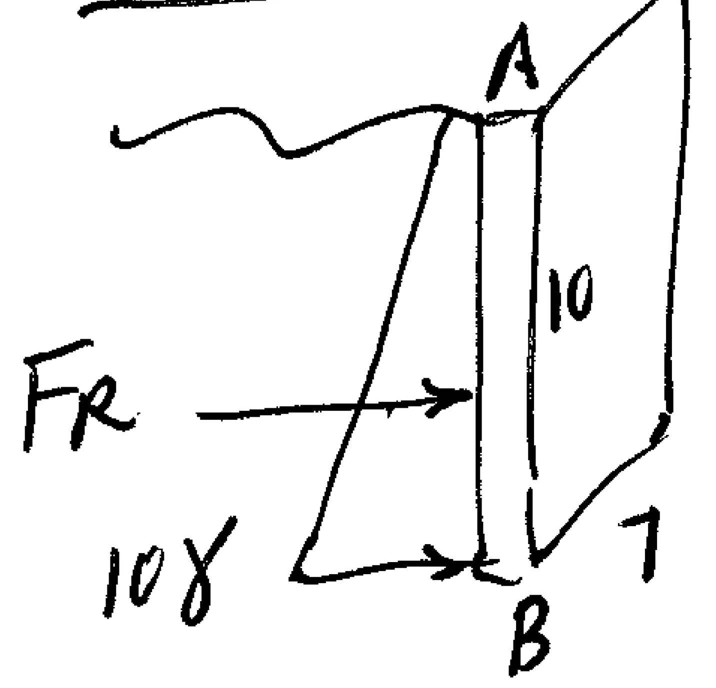
$SG = \frac{65}{62.4} = 1.042$

Determine:

① F_R on AB

② F_R on BC

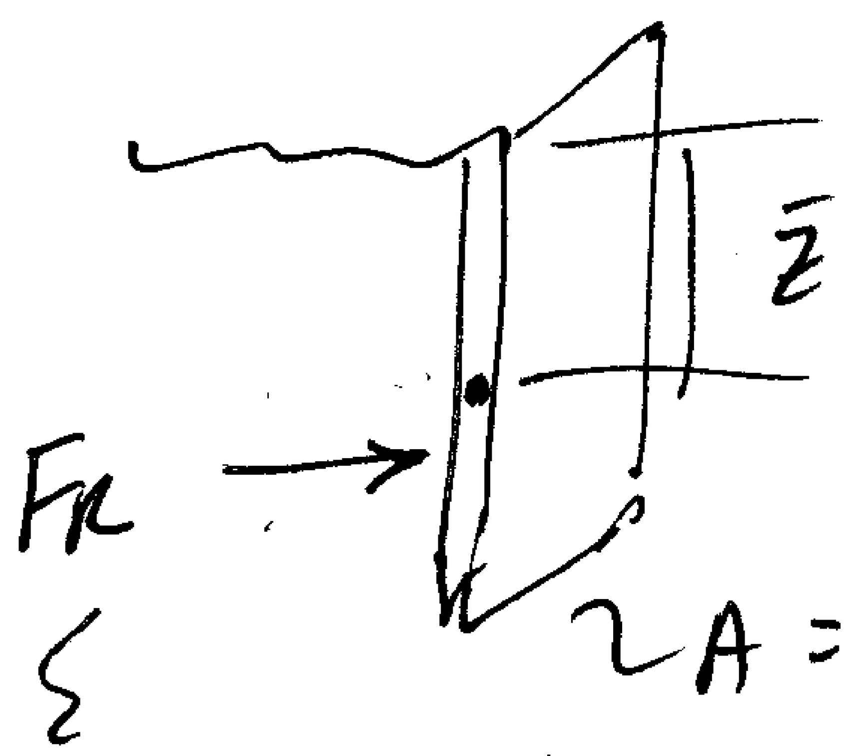
① F_R on AB



(a) $\boxed{F_R = \frac{1}{2} b h w}$ $\gamma = 65 \ lb/ft^3$

$= \frac{1}{2}(10\gamma)(10)(7) = \boxed{22.75 \text{ kip} = F_R}$

(b) $\boxed{F_R = \gamma \cdot \bar{z} \cdot A}$



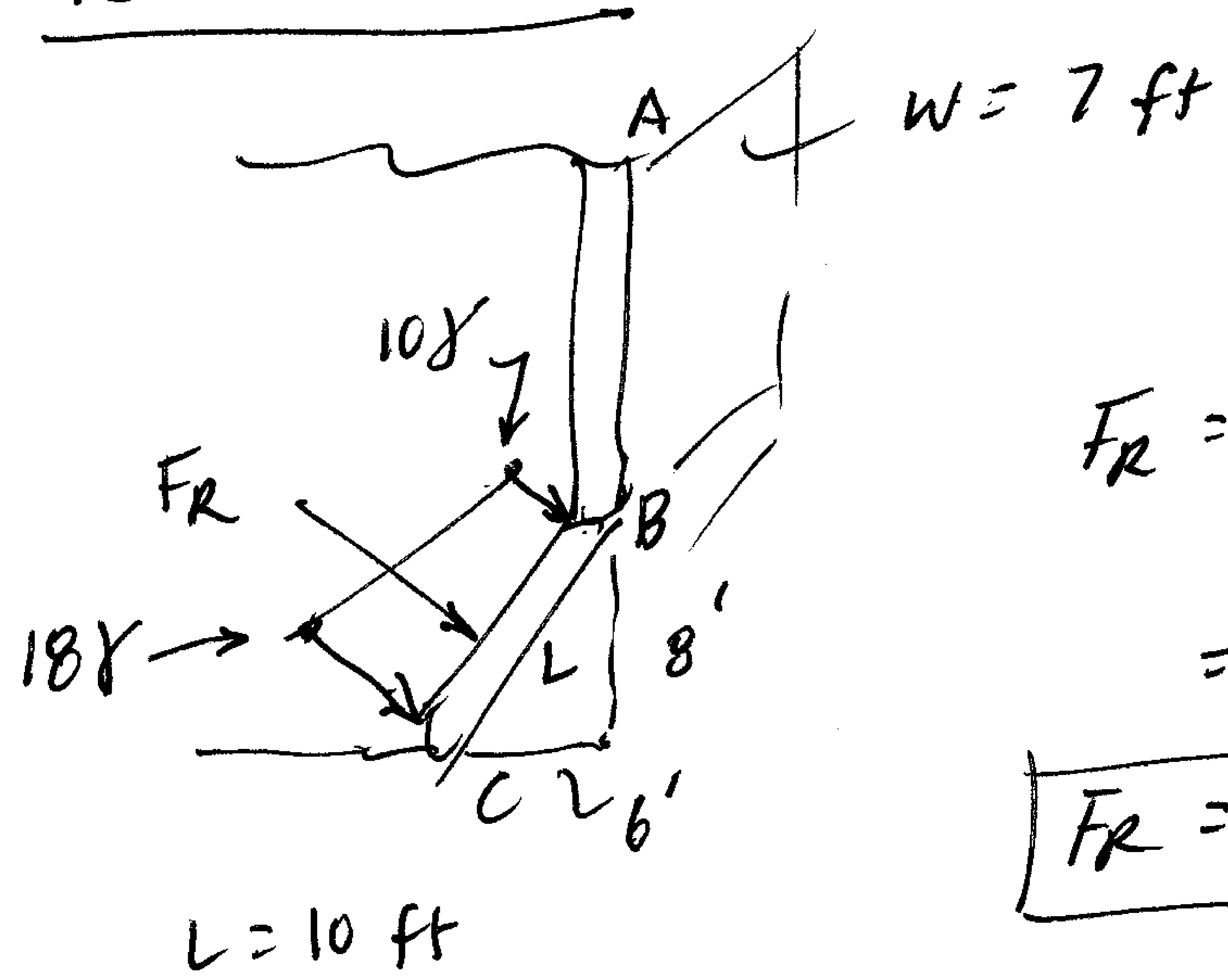
$= (65)(5)(70)$

$\boxed{F_R = 22.75 \text{ kip}}$

F_R acts at $\frac{10}{3} = 3.33$ ft up from B.

Does not act at $\bar{z}$!

② F_R on BC



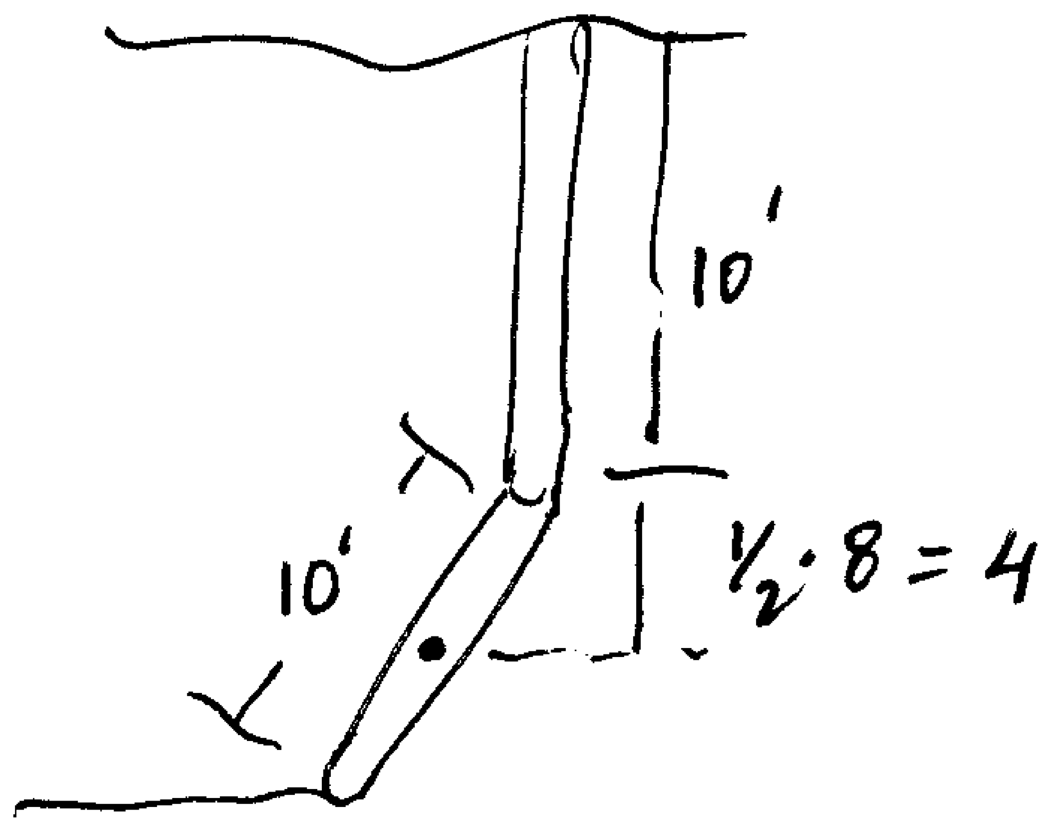
(a) F_R = Trapezoid ...

$$F_R = \left(\frac{10\gamma + 18\gamma}{2}\right)(10')(7')$$

$$= 14\gamma \cdot 70 = 980\,\gamma$$ 65

$$\boxed{F_R = 63.7 \text{ kip}}$$

(b) $\boxed{F_R = \gamma \cdot \bar{z} \cdot A}$



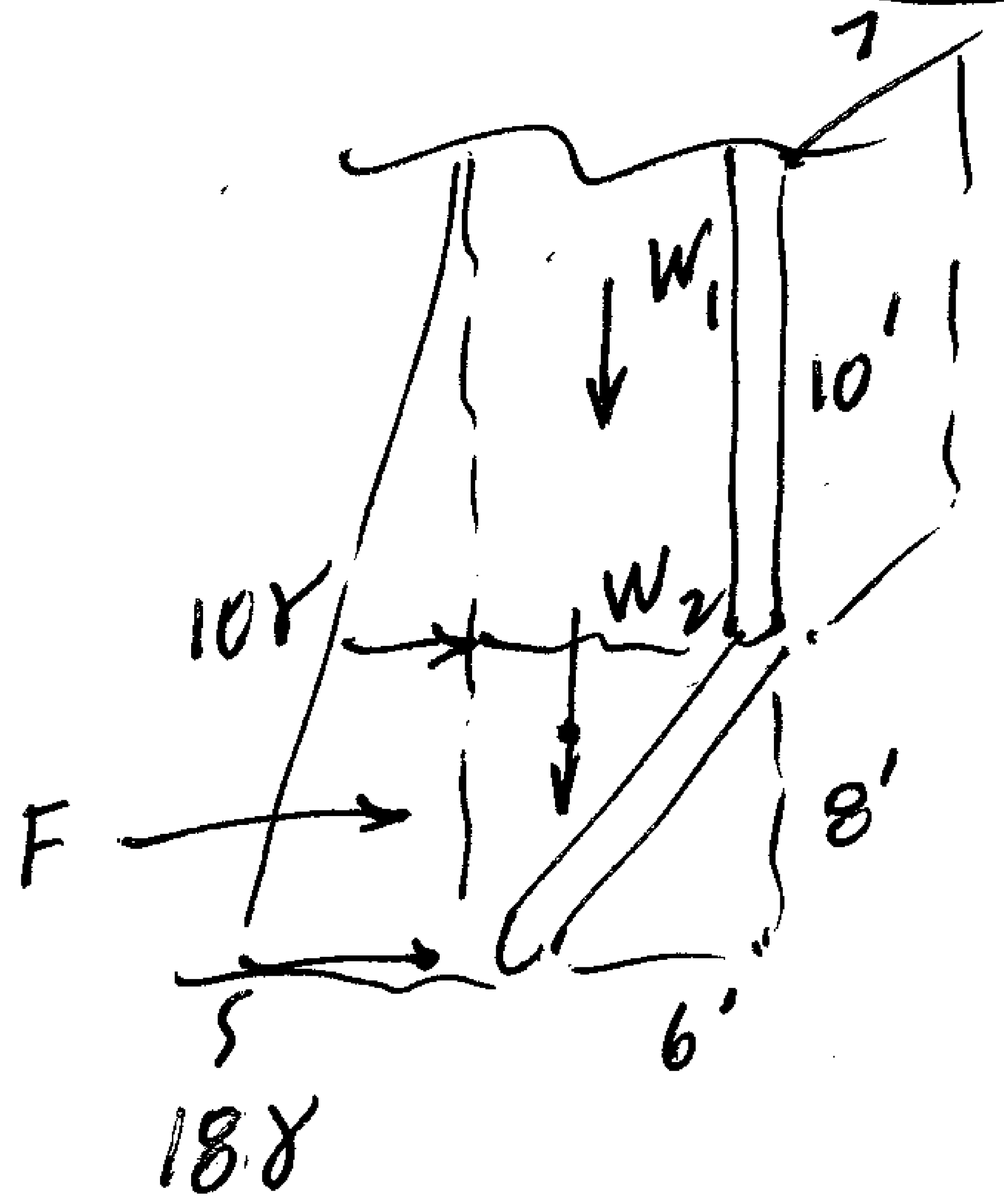
$\bar{z} = 10 + 4 = 14'$

$A = 10(7) = 70$

$$F_R = 65(14)(70) = 980\,\gamma$$

$$\boxed{F_R = 63.7 \text{ kip}}$$

(c) Use $\boxed{\vec{F}_R = \vec{F}_H + \vec{F}_V}$



$$F_V = W_1 + W_2$$

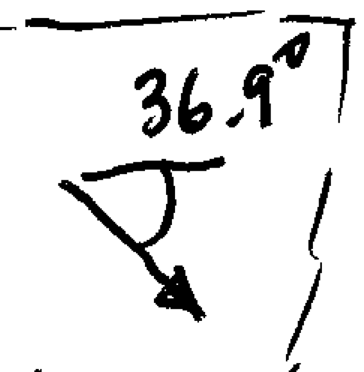
$$= (10)(6)(7)\gamma + \tfrac{1}{2}(6)(8)(7)\gamma$$ 65

$$= 420\gamma + 168\gamma = 588\,\gamma$$

$$\boxed{F_V = 38.22 \text{ kip}} \downarrow$$

$$F_H = \left(\frac{10\gamma + 18\gamma}{2}\right)(8)(7) = 784\,\gamma$$

$$\boxed{F_H = 50.96 \text{ kip}} \rightarrow$$



$$\vec{F}_R = \vec{F}_H + \vec{F}_V = \left[50.96\,\hat{i} - 38.22\,\hat{j}\right] = \boxed{63.7 \text{ kip}}$$